Supply Chain ML Notebook Manual

Introduction:

The Supply Chain ML Predictor uses a dataset to train a model that estimates late-delivery risk and delay duration. The notebook provides two dataset options: download the provided dataset or upload a custom dataset.

Things we will need before running the notebook:

- The notebook file: **SCM+ML_project.ipynb**.
- Google Colab or a compatible Jupyter environment.
- An internet connection for the download option, or your own dataset file for the custom upload option.

Theory/Procedure:

1. Choose the dataset option

In the dataset-selection cell, choose one of the two available options: download the provided DataCo dataset from the configured GitHub link, or upload your own dataset file when prompted.

2. Upload a custom dataset (if selected)

Upload an Excel file (.xlsx) or CSV file (.csv). Keep the column names exactly as expected by the notebook. The custom file must include the fields used for the model, including **Type**, **Category Name**, **Customer Segment**, **Market**, **Order Region**, **Shipping Mode**, **Days for shipment (scheduled)**, **Order Item Total**, **Order Item Quantity**, **Days for shipping (real)**, and **Late_delivery_risk**. Do not rename these columns.

3. Load and inspect the dataset

Run the dataset-loading cell after choosing the download or upload option. The notebook loads the data and prints the row and column counts. If a file-name error appears, check the exact uploaded filename and the filename used in the loading cell.

4. Prepare the data

The notebook selects the required fields, removes rows with missing required values, and prepares the categorical and numerical features. It calculates **Delay_Days** from actual shipping days minus scheduled shipping days, then splits the data for training and testing.

5. Train and evaluate the model

Run the cells that build and train the neural network, then run the evaluation cell. Review the classification and delay-duration metrics shown by the notebook. Results depend on the selected dataset and training run.

6. Save the model files

The save step creates **supply_chain_multi_output_model.h5** and **preprocessor_multi.pkl** if it finishes successfully. Keep both files if you plan to use the trained model outside the notebook.

7. Use the prediction form

In the final interactive form, enter the shipment details using the available dropdowns and numeric controls: Type, Category Name, Customer Segment, Market, Order Region, Shipping Mode, scheduled shipping days, Order Item Total, and Order Item Quantity. Click **Predict Delay Risk & Duration** to view the result.

8. Run all cells and verify the output

Run all notebook cells from beginning to end in Google Colab. Make sure every cell executes successfully without errors, both dataset options work (download and custom .xlsx/.csv upload), the model trains and evaluation completes, the model files are saved, and the final prediction form produces a result. If any cell fails, fix the issue and rerun the affected cells before considering the notebook complete.